Data Science 500 Assignment

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QUESTION	EXAMINER MARKS	MODERATOR MARKS	REMARKS

ASSIGNMENT INSTRUCTIONS

Please tick each box to confirm completion.

Use Times New Roman font, size 12, with 1.5 line spacing throughout the document. Apply Harvard Referencing Style for all citations and references.

For essay-style assignments, please include the following sections:

Table of Contents Introduction

Main Body (with relevant subheadings) Conclusion

References

Submit the assignment in PDF format on Moodle. Use the specified cover page provided.

Include a signed declaration of originality.

DECLARATION OF ORIGINALITY:

I hereby declare that this assignment is my own work and has not been copied from any other source except where due acknowledgment is made. I affirm that all sources used have been properly cited and that this submission complies with the institution's policies on academic integrity and plagiarism.

Student Signature: M.mp Date: 23/10/2026

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Introduction

The aim of this project is to analyse book prices from the website books.toscrape.com. I will be using Python to collect data from the website. I used BeautifulSoup to scrape the website and Pandas to save the data. I collected 1000 books from the website. For each book, I collected the title, price, rating, availability and category. I saved all the data in a file called books_dataset.csv.

After collecting the data, I made 5 graphs to understand the data better. I also used machine learning to predict the price of a book based on its rating and category. This project will help the publishing analytics team to understand how the price of a book changes with its rating and its category.

Web Scraping and Machine Learning Analysis of Book Prices.

```
import requests
from bs4 import BeautifulSoup
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import time
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error
from sklearn.metrics import r2_score

books_list = []

print("Starting to scrape books...")
print("Please wait, it will take a few minutes")

rating_dict = {}
rating_dict["One"] = 1
rating_dict["Two"] = 2
rating_dict["Three"] = 3
rating_dict["Four"] = 4
rating_dict["Five"] = 5

my_headers = {}
my_headers["User-Agent"] = "Mozilla/5.0"

page_number = 1
while page_number <= 50:
    my_link = "https://books.toscrape.com/catalogue/page-" + str(page_number) + ".html"

    try:
        my_page = requests.get(my_link, headers=my_headers, timeout=10)
    except:
        print("Internet problem on page", page_number, "trying again")
        time.sleep(3)
        try:
            my_page = requests.get(my_link, headers=my_headers, timeout=10)
        except:
            print("Still problem, skipping page", page_number)
            page_number = page_number + 1
            continue

    my_soup = BeautifulSoup(my_page.text, "html.parser")
    books_on_page = my_soup.find_all("article", class_="product_pod")
```

```
for one_book in books_on_page:
    title = one_book.h3.a["title"]

    price_tag = one_book.find("p", class_="price_color")
    price_text = price_tag.text
    price_text = price_text.replace("£", "")
    price_text = price_text.replace("Â", "")
    price_text = price_text.strip()
    price = float(price_text)

    rating_tag = one_book.find("p", class_="star-rating")
    rating_word = rating_tag["class"][1]
    rating = rating_dict[rating_word]

    stock_tag = one_book.find("p", class_="instock availability")
    stock_text = stock_tag.text
    stock_text = stock_text.strip()
    if "In stock" in stock_text:
        in_stock = True
    else:
        in_stock = False
    try:
        book_link = one_book.h3.a["href"]
        full_book_link = "https://books.toscrape.com/catalogue/" + book_link
        book_page = requests.get(full_book_link, headers=my_headers, timeout=10)
        book_soup = BeautifulSoup(book_page.text, "html.parser")
        breadcrumb = book_soup.find("ul", class_="breadcrumb")
        breadcrumb_items = breadcrumb.find_all("li")
        category = breadcrumb_items[2].text
        category = category.strip()
        time.sleep(0.3)
    except:
        print("Could not get category for", title, "- putting Unknown")
        category = "Unknown"

    one_book_data = {}
    one_book_data["title"] = title
    one_book_data["price"] = price
    one_book_data["rating"] = rating
    one_book_data["rating_word"] = rating_word
    one_book_data["in_stock"] = in_stock
    one_book_data["category"] = category

    books_list.append(one_book_data)

print("Finished page", page_number, "Total books now:", len(books_list))
page_number = page_number + 1
```

```
df = pd.DataFrame(books_list)
df = df.dropna()
df = df.drop_duplicates()
print("Total books after cleaning:", len(df))
print(df.head())
df.to_csv("books_dataset.csv", index=False)
print("Saved csv file")

plt.figure(figsize=(8, 5))
sns.barplot(x="rating", y="price", data=df)
plt.title("Average Price by Rating")
plt.savefig("price_by_rating.png")
plt.close()
print("Made graph 1")

category_price = df.groupby("category")["price"].mean()
category_price = category_price.sort_values(ascending=False)
top_10 = category_price.head(10)
plt.figure(figsize=(10, 5))
top_10.plot(kind="bar")
plt.title("Average Price by Category - Top 10")
plt.savefig("price_by_category.png")
plt.close()
print("Made graph 2")

plt.figure(figsize=(8, 5))
sns.histplot(df["price"], bins=20)
plt.title("Price Distribution")
plt.savefig("price_distribution.png")
plt.close()
print("Made graph 3")

plt.figure(figsize=(8, 5))
sns.countplot(x="rating", data=df)
plt.title("Number of Books by Rating")
plt.savefig("rating_count.png")
plt.close()
print("Made graph 4")

X = df[["rating"]]
y = df["price"]
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
my_model = LinearRegression()
my_model.fit(X_train, y_train)
y_pred = my_model.predict(X_test)
my_mse = mean_squared_error(y_test, y_pred)
my_r2 = r2_score(y_test, y_pred)
print("MSE:", my_mse)
```

```
print("R2 Score:", my_r2)

plt.figure(figsize=(8, 5))
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Price")
plt.ylabel("Predicted Price")
plt.title("Actual vs Predicted Price")
plt.savefig("actual_vs_predicted.png")
plt.close()
print("Made graph 5")

print("ALL DONE")
```

```
C:\Users\mpho\.local\bin\uv.exe run C:/Users/mpho/PycharmProjects/PythonProject10/.venv/Scripts/python.exe C:\Users\mpho\PycharmProjects\PythonProject10\main.py
Starting to scrape books...
Please wait, it will take a few minutes
Finished page 1 Total books now: 20
Finished page 2 Total books now: 40
Finished page 3 Total books now: 60
Finished page 4 Total books now: 80
Finished page 5 Total books now: 100
Finished page 6 Total books now: 120
Finished page 7 Total books now: 140
Finished page 8 Total books now: 160
Finished page 9 Total books now: 180
Finished page 10 Total books now: 200
Finished page 11 Total books now: 220
Finished page 12 Total books now: 240
Finished page 13 Total books now: 260
Finished page 14 Total books now: 280
Finished page 15 Total books now: 300
Finished page 16 Total books now: 320
Finished page 17 Total books now: 340
Finished page 18 Total books now: 360
Finished page 19 Total books now: 380
Finished page 20 Total books now: 400
Finished page 21 Total books now: 420
```

```
Finished page 22 Total books now: 440
Finished page 23 Total books now: 460
Finished page 24 Total books now: 480
Finished page 25 Total books now: 500
Finished page 26 Total books now: 520
Finished page 27 Total books now: 540
Could not get category for A Short History of Nearly Everything - putting Unknown
Finished page 28 Total books now: 560
Finished page 29 Total books now: 580
Finished page 30 Total books now: 600
Finished page 31 Total books now: 620
Could not get category for Evicted: Poverty and Profit in the American City - putting Unknown
Finished page 32 Total books now: 640
Finished page 33 Total books now: 660
Finished page 34 Total books now: 680
Could not get category for Animal Farm - putting Unknown
Could not get category for A Walk to Remember - putting Unknown
Could not get category for A New Earth: Awakening to Your Life's Purpose - putting Unknown
Could not get category for A History of God: The 4,000-Year Quest of Judaism, Christianity, and Islam - putting Unknown
Could not get category for 'Salem's Lot - putting Unknown
Could not get category for Zero History (Blue Ant #3) - putting Unknown
Could not get category for Wuthering Heights - putting Unknown
Finished page 35 Total books now: 700
Finished page 36 Total books now: 720
```

```
Could not get category for The Nanny Diaries (Nanny #1) - putting Unknown
Could not get category for The Name of God is Mercy - putting Unknown
Finished page 37 Total books now: 740
Could not get category for The Lean Startup: How Today's Entrepreneurs Use Continuous Innovation to Create Radically Successful Businesses - putting Unknown
Could not get category for The Last Painting of Sara de Vos - putting Unknown
Could not get category for The Land of 10,000 Madonnas - putting Unknown
Could not get category for The Infinities - putting Unknown
Could not get category for The Husband's Secret - putting Unknown
Could not get category for The Guernsey Literary and Potato Peel Pie Society - putting Unknown
Could not get category for The Goldfinch - putting Unknown
Could not get category for The Giver (The Giver Quartet #1) - putting Unknown
Could not get category for The Girl with All the Gifts - putting Unknown
Could not get category for The Girl Who Kicked the Hornet's Nest (Millennium Trilogy #3) - putting Unknown
Could not get category for The Exiled - putting Unknown
Could not get category for The End of Faith: Religion, Terror, and the Future of Reason - putting Unknown
Finished page 38 Total books now: 760
Finished page 39 Total books now: 780
Finished page 40 Total books now: 800
Finished page 41 Total books now: 820
Finished page 42 Total books now: 840
Finished page 43 Total books now: 860
```

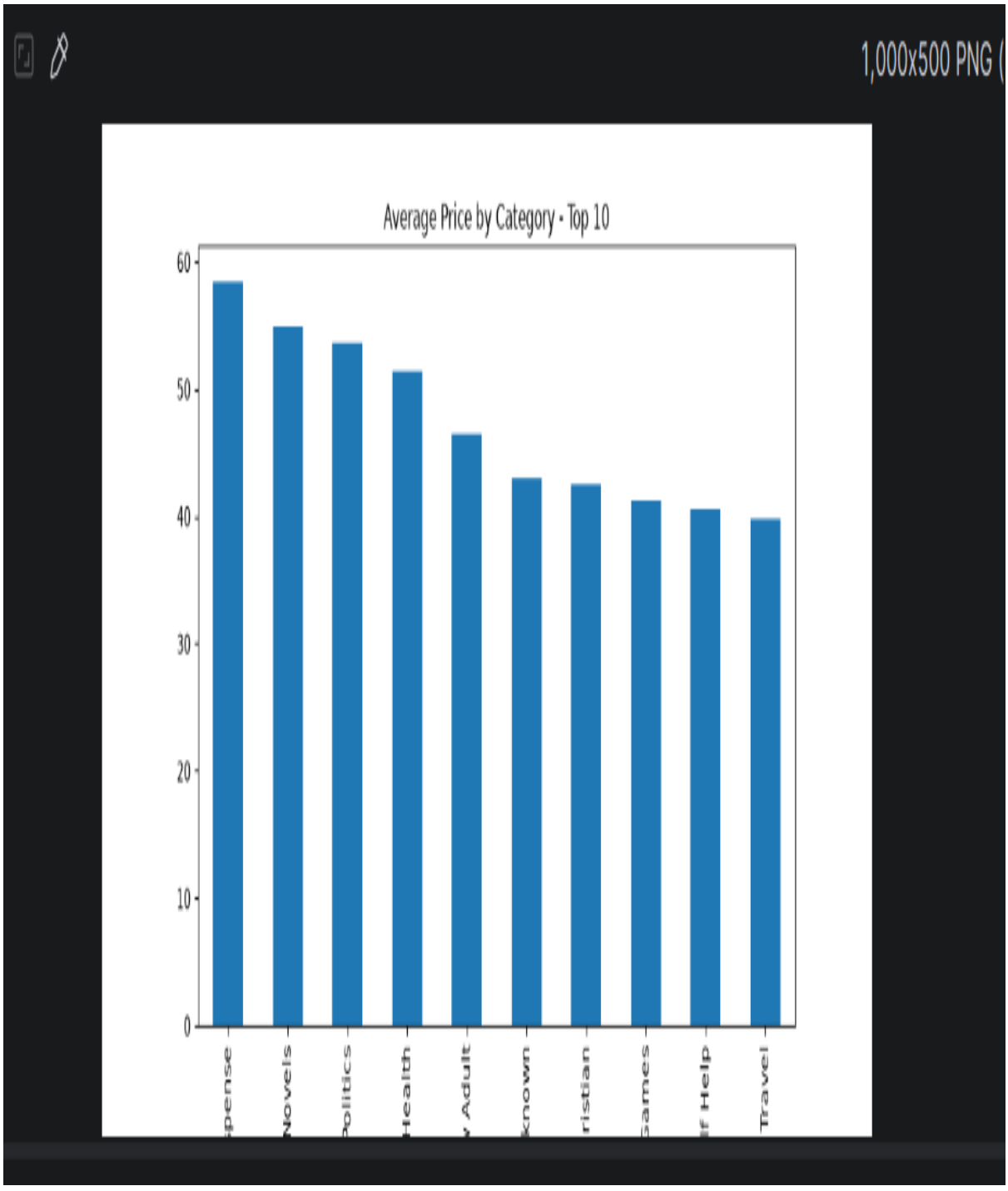
```
Finished page 44 Total books now: 880
Finished page 45 Total books now: 900
Finished page 46 Total books now: 920
Finished page 47 Total books now: 940
Finished page 48 Total books now: 960
Finished page 49 Total books now: 980
Could not get category for Alice in Wonderland (Alice's Adventures in Wonderland #1) - putting Unknown
Could not get category for Ajin: Demi-Human, Volume 1 (Ajin: Demi-Human #1) - putting Unknown
Finished page 50 Total books now: 1000
Total books after cleaning: 1000
                               title  ...            category
0                 A Light in the Attic  ...              Poetry
1                   Tipping the Velvet  ...  Historical Fiction
2                           Soumission  ...             Fiction
3                        Sharp Objects  ...             Mystery
4  Sapiens: A Brief History of Humankind  ...             History

[5 rows x 6 columns]
Saved csv file
Made graph 1
Made graph 2
Made graph 3
Made graph 4
MSE: 211.4538481028864
```

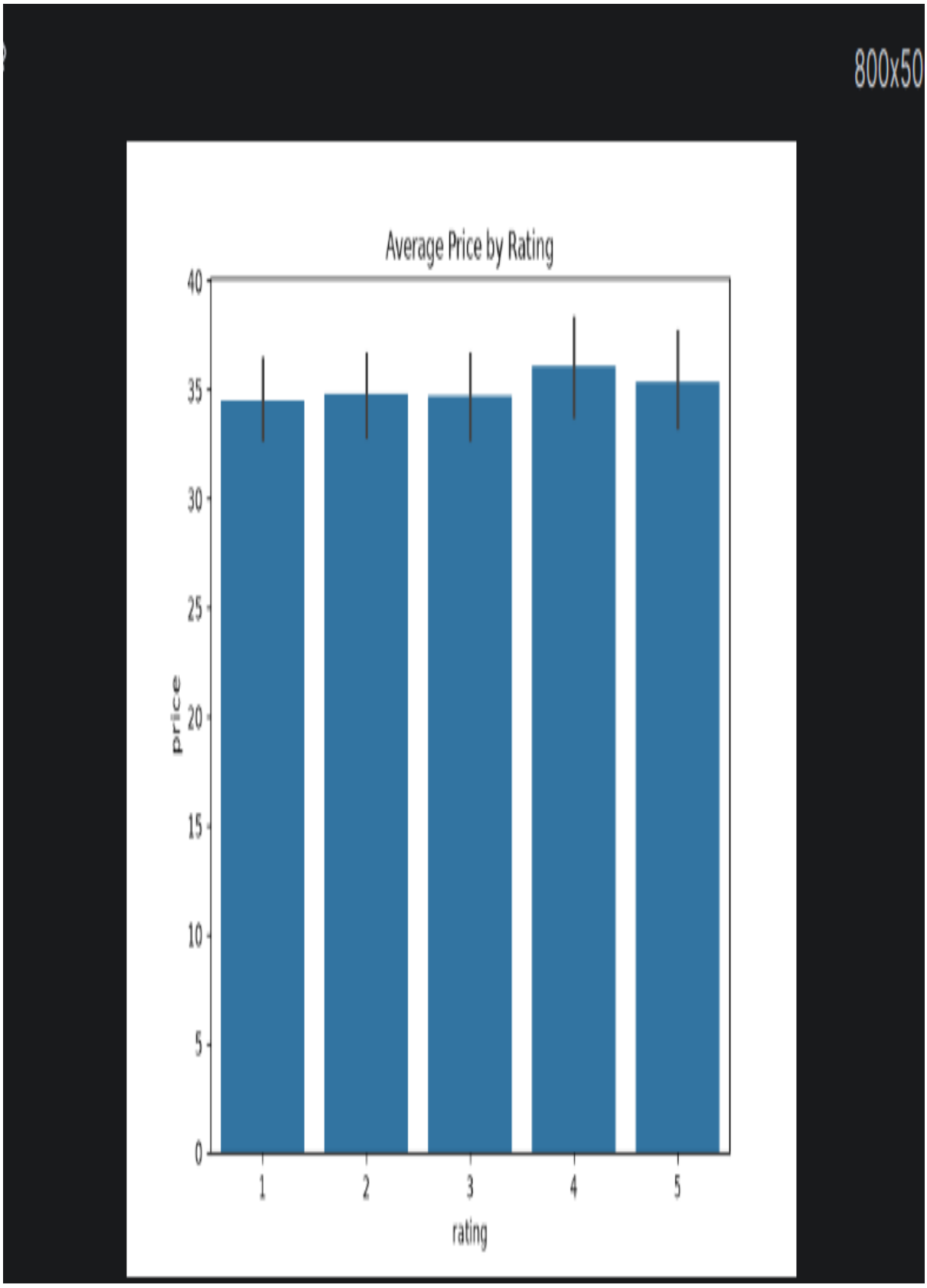
```
MSE: 211.4538481028864
R2 Score: -0.006618384017733847
Made graph 5
ALL DONE

Process finished with exit code 0
```

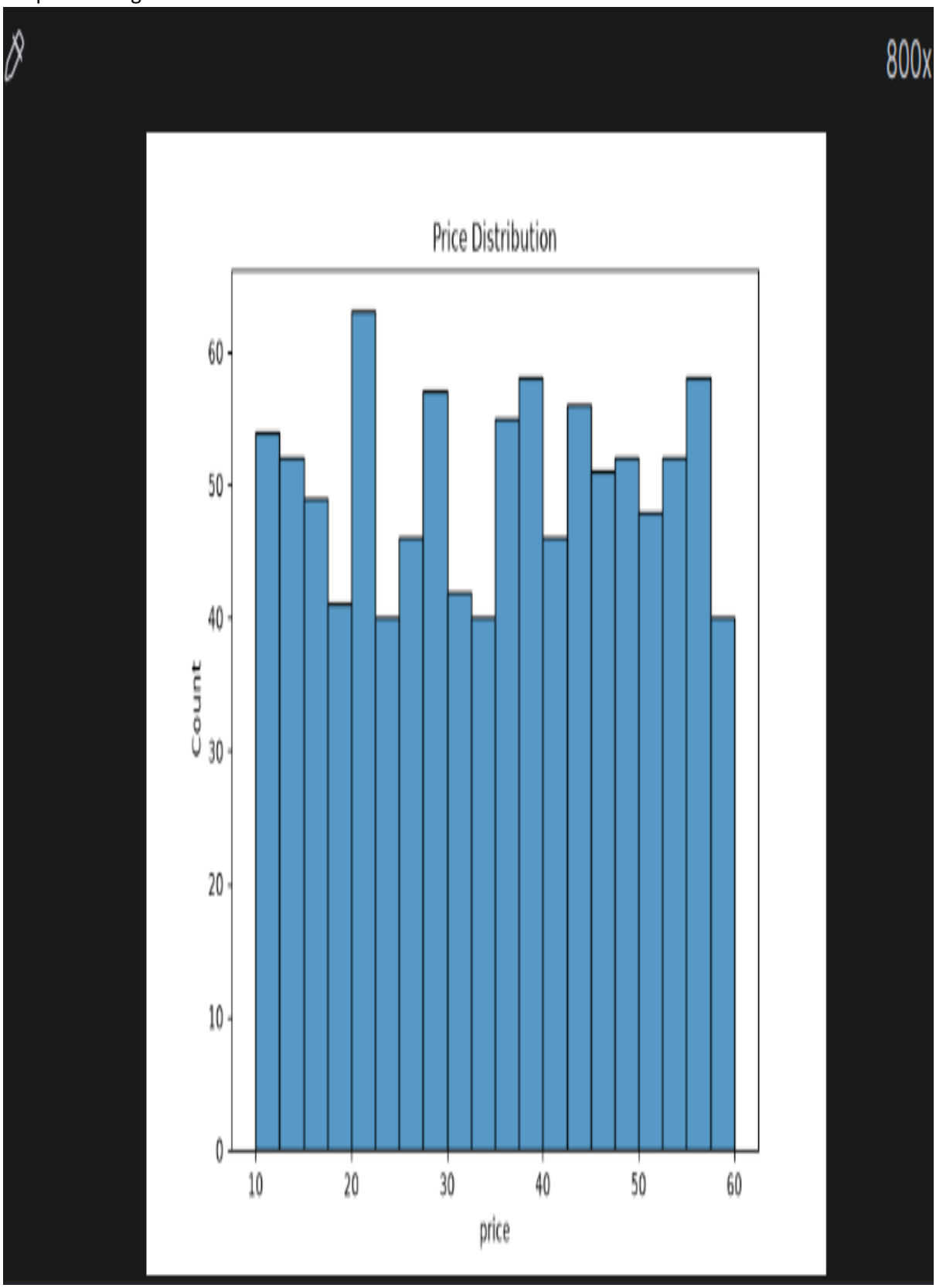
This is the graph of Average Price by Category



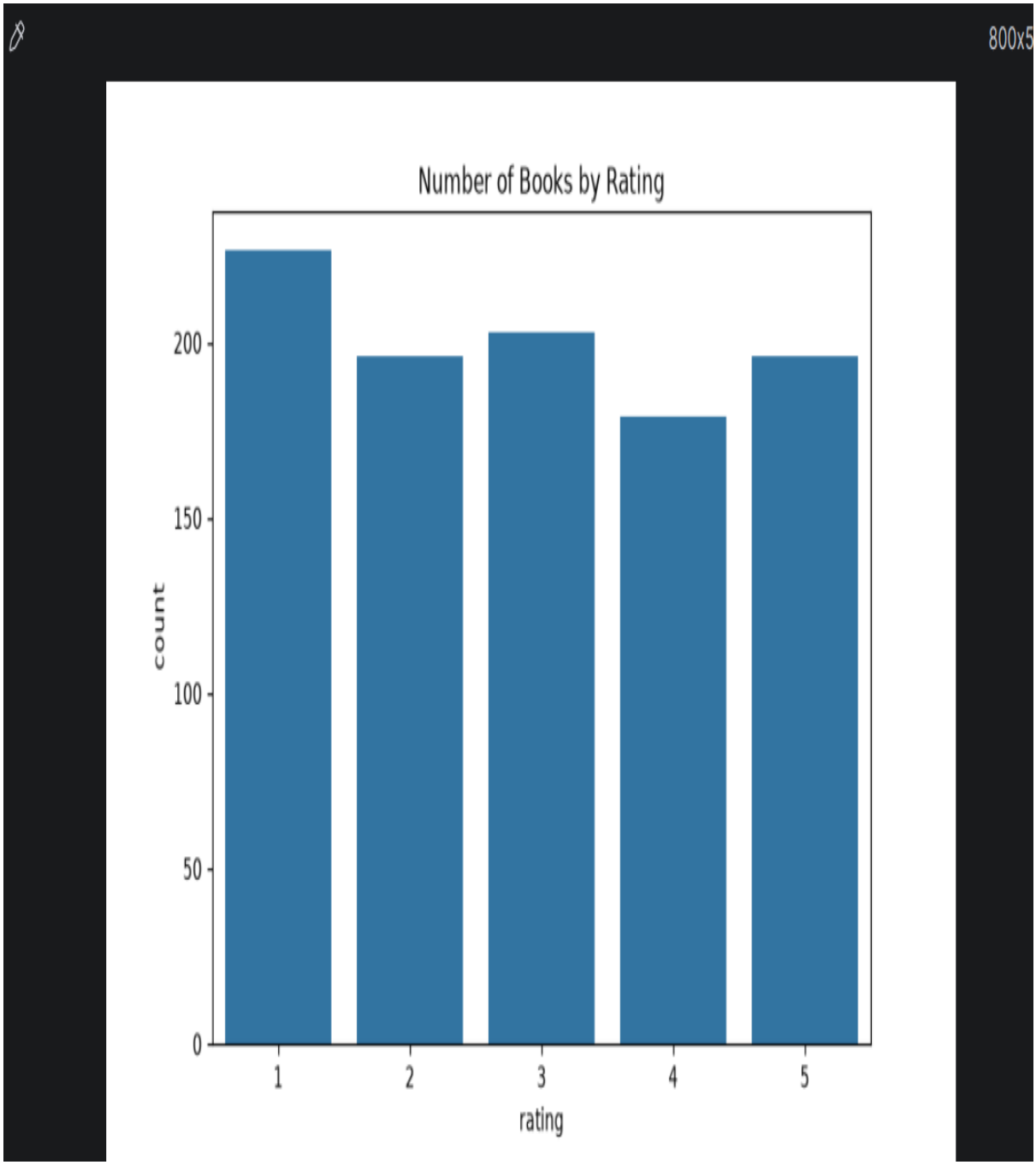
Graph showing the Average Price by Rating



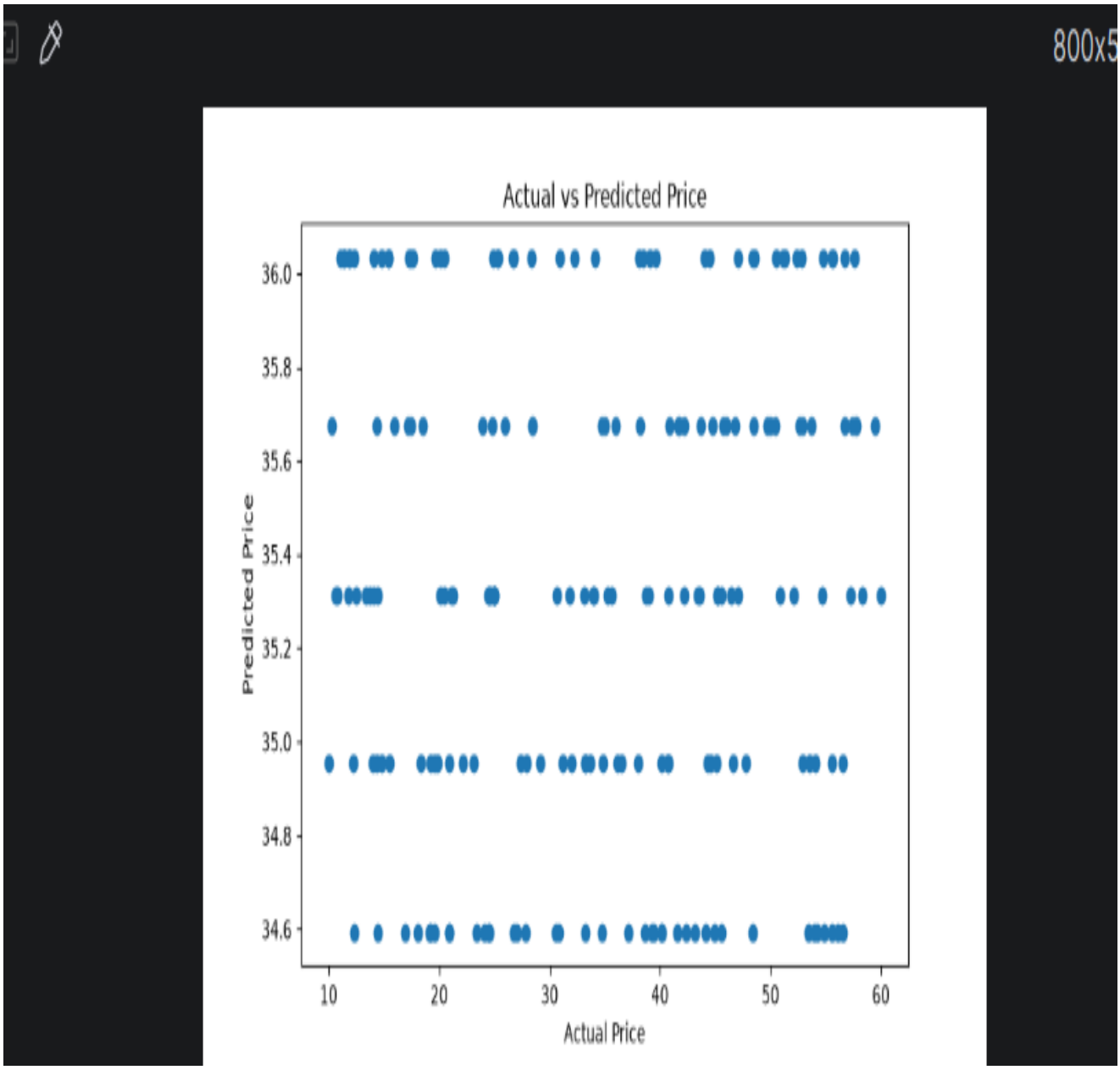
Graph showing Price Distribution VS Count



Graph showing Number of Books by Rating



Graph showing Actual Price VS Predicted Price



Conclusion

In this project, I successfully collected 1000 books from books.toscrape.com and saved them in books_dataset.csv. From Graph 1, I found that the average price changes with the rating of the book. From Graph 2, I found that some categories of books are more expensive than other categories. From Graph 3, I found that most books cost between £20 and £40. From Graph 4, I found that most books have a rating of 1, 2 or 3 stars.

From Graph 5, I used machine learning to predict the price of a book. My model gave an MSE and an R2 Score. This shows that rating and category alone do not perfectly predict the price of a book, because other factors also affect the price. This project helped me to learn how to scrape a website, clean data, make graphs and use machine learning in Python.

References

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